



Department of Computer Science and Engineering
National Institute of Technology, Rourkela
End Semester Examination (Autumn 2025 - 26)
MTech 1st Semester
Sub: Foundations of Computer Security (CS - 6301)
Time: 3 Hrs; FM: 50

Answer **ALL** questions.

1. (a) Find the result of $2001^{35} \bmod 2000$ using the square-and-multiply method. (5,CO3,BL2)
(b) Solve $x^2 \equiv 4 \bmod 14$ using quadratic residue. (5,CO2,BL3)
2. (a) Show the first eight words of the key expansion in AES for a 128-bit key of all ones. [The key consists of all 1's] (5,CO4,BL3)
(b) Given the plain text **000102030405060708090A0B0C0D0E0F** and the key **01010101010101010101010101010101**. Show the original content of the state and the value of the state after initial AddRoundKey in AES. (5,CO4,BL3)
3. (a) The plaintext and key in DES is same and is given as **(0123456789ABCDEF)₁₆**. Derive the first round key K_1 , and L_0 and R_0 . Compute $f(R_0) \text{ XOR } K_1$. Use the expansion P-box given in Table 1. (5,CO4,BL3)
(b) Use the extended Euclidean algorithm to find the inverse of $x^4 + x^3 + 1$ in $GF(2^5)$ using the modulus $x^5 + x^2 + 1$. (5,CO3 BL3)
4. (a) Use Vignere cipher with keyword **HEALTH** to encipher the message "life is full of surprise". (5,CO3 BL2)
(b) The encryption key in a transposition cipher is **(3, 2, 6, 1, 5, 4)**. Find the decryption key. (5,CO2,BL3)
5. Answer the following the questions: [5x2 = 10]
 - a. Find the n-bit word represented by the polynomial $x^2 + 1$ in $GF(2^4)$. (CO3,BL2)
 - b. Define an abelian group. (CO3,BL1)
 - c. What is the difference between SubBytes and SubWord in AES. (CO2,BL1)
 - d. Distinguish between diffusion and confusion. (CO3,BL1)
 - e. Find the value of $\phi(29)$. (CO4,BL2)

Table 1

32	01	02	03	04	05
04	05	05	06	08	09
08	09	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	01